

Navigation Techniques

in JSP & Servlets

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Introduction to Page Navigation

- Page navigation in JSP and Servlets involves directing users between different pages of a web application. There are multiple techniques available, each with its own use cases.

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1. RequestDispatcher (Forward & Include)

- RequestDispatcher is used for internal forwarding and including content from another resource.
- Forwarding (Forward)
 - Transfers request processing to another resource (JSP/Servlet).
 - The client (browser) remains unaware of this transfer.
 - URL in the browser remains the same.
- Example (Forwarding using RequestDispatcher in Servlet):


```
RequestDispatcher dispatcher =
request.getRequestDispatcher("nextPage.jsp");
dispatcher.forward(request, response);
```

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forward action

- Lets you abort execution of the current page and transfer the request to another page:


```
<jsp:forward page="myOtherPage.jsp">
<jsp:param name="newParName" value="newParValue"/>
</jsp:forward>

...
<% String dest = "/myJspPages/" + someVar; %>
<jsp:forward page="<%=dest%>">
<jsp:param name="newParName" value="newParValue"/>
</jsp:forward>
```

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- If the previous logic done using pure java code:

```
... <%  
String dest = "/myJspPages/" + someVar;  
RequestDispatcher rd = application.getRequestDispatcher(dest + "?newParName=newParValue");  
rd.forward(request, response);  
%>
```

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- Including (Include)
 - Includes content from another resource without stopping current processing.
 - Useful for including headers, footers, or common UI components.
- Example (Including a JSP Page in another JSP):
`<jsp:include page="header.jsp" />`

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2. Send Redirect

- Redirects the client to a new URL.
- The browser makes a new request, meaning the URL changes.
- Useful for redirecting to external websites or handling post-redirect-get patterns.
- Example (Redirecting in Servlet):
 - `response.sendRedirect("https://www.example.com");`
- Example (Redirecting within the application):
 - `response.sendRedirect("home.jsp");`

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3. Hyperlink Navigation

- Simple navigation using `<a>` tags in JSP.
- Used when direct user interaction is needed.
- Example (Link to another JSP page):
 - `<a href="contact.jsp">Contact Us</a>`
- Example (Passing Parameters in URL):
 - `<a href="profile.jsp?userId=123">View Profile</a>`

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4. Form Submission Navigation

- Navigation based on user input.
- Data is sent to a servlet or JSP via POST or GET methods.

- Example (Form sending data to Servlet):

```
<form action="LoginServlet" method="post">  
Username: <input type="text" name="username" />  
Password: <input type="password" name="password" />  
<input type="submit" value="Login" />  
</form>
```

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5. JavaScript-Based Navigation

- Allows client-side redirection using JavaScript.
- Used for dynamic behavior or time-based redirection.

- Example (Redirect using JavaScript in JSP):

```
<script> window.location.href = "home.jsp"; </script>
```

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Point to remember!

- **Tomcat clears the output buffer upon executing the forward action.**
 - Therefore, the HTML code generated up to that point by the current page is lost.
- But if the current page has already filled the response buffer by the time it is aborted with forward, that part of the response will have already left the server.
- This will probably result in a bad page sent to the client.

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Point to remember!

- `<jsp:include page="TopMenu.jsp" flush="true"/>`
- The flush attribute (default false) ensures that the HTML generated so far by the including page is sent to the client before executing the included page.

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Point to remember!

- 1- Forward vs 2 – include
- With `jsp:forward`, Tomcat clears the output buffer upon executing the forward action.
- With `jsp:include` Tomcat doesn't clear the output buffer when it executes that action.

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Remember Again!

- *RequestDispatcher* is used when you want to redirect from the currently loading page to a another page automatically.
- The new page must contains a complete structure of a valid HTML document.

```
<htm>
<head><title></title></head>
<body></body>
</html>
```

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Remember Again!

- *Jsp:include* is used when you want to load some other page into the currently loading page.
- Commonly used to load copyright statement, banner, menus etc.
- The sum of included page + including page must form a complete structure of a valid HTML document.

```
<html>
<head><title></title></head>
<body></body>
</html>
```

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Comparing Navigation Methods

Method	URL Change?	Server Request Type	Use Case
RequestDispatcher (Forward)	No	Internal Request	Passing control within the server
RequestDispatcher (Include)	No	Internal Request	Reusing UI components like headers/footers
Send Redirect	Yes	New Request	Redirecting to external pages or after a form submission
Hyperlink (<a> tag)	Yes	New Request	Simple user navigation
Form Submission	Yes	New Request	Handling user input
JavaScript Navigation	Yes	New Request	Dynamic redirection

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Best Practices for Page Navigation

- Use forwarding (RequestDispatcher) for internal requests to maintain request attributes.
- Use redirect (sendRedirect) when redirecting to a different domain or after a POST request.
- Use form-based navigation for sending data securely.
- Minimize JavaScript-based navigation for SEO and accessibility.
- Keep URLs clean and readable for better user experience.